Tianyi Yu

CONTACT INFORMATION

LACIM
Université du Québec à Montréal
Montréal, QC, Canada

E-mail: yu.tianyi@uqam.ca
Website: https://tianyi-math.github.io/

EDUCATION

University of California, San Diego, La Jolla, CA, USA 2020 - 2024
PhD in Mathematics, Advisor: Brendon Rhoades

University of California, San Diego, La Jolla, CA, USA 2016 - 2020
Bachelor of Science in Mathematics & Bachelor of Science in Computer Science

EMPLOYMENT

University of Illinois Urbana-Champaign, Urbana, IL, USA Jan. 2027 –
J. L. Doob Postdoctoral Fellow,
Mentor: Alex Yong

Laboratoire d'Algèbre, de Combinatoire et d'Informatique Mathématique,
Montréal, QC, Canada Oct. 2024 - Dec. 2026
CRM-ISM Postdoctoral Fellow,
Mentors: François Bergeron, Jake Levinson, Alejandro Morales, Hugh Thomas, and Franco Saliola

RESEARCH INTERESTS

Algebraic combinatorics: Schubert calculus, Lascoux polynomials, Grothendieck polynomials, bumpless pipedreams, crystals, and coinvariant algebras.

PREPRINTS

[1] *A positive combinatorial formula for the double Edelman–Greene coefficients*, with Chou, C. Preprint arXiv:2512.12185

PUBLICATIONS

[2] *Grothendieck positivity for normal square root crystals*, with Marberg, E. & Tong, K. H. Advances in Mathematics (2026).

[3] *Tableau formula for vexillary double Edelman–Greene coefficients*, with Gregory, A. & Hamaker, Z. Selecta Mathematica (accepted); arXiv:2412.20615.

[4] *Marked bumpless pipedreams and compatible pairs*, with Huang, D. & Shimozono, M. Combinatorial Theory (accepted).

[5] *Embedding bumpless pipedreams as Bruhat chains*. International Mathematics Research Notices (2025).

[6] *Grothendieck polynomials of inverse fireworks permutations*, with Chou, C. European Journal of Combinatorics (2025).

[7] *Lascoux expansion of the product of a Lascoux and a stable Grothendieck*, with Orelowitz, G. Combinatorial Theory (accepted); arXiv:2312.01647.

[8] *Constructing maximal pipedreams of double Grothendieck polynomials*, with Chou, C. The Electronic Journal of Combinatorics (2024).

[9] *Constructing a Gröbner basis of Griffin's ideal*.
Mathematische Zeitschrift (2024).

[10] *Connection between Schubert polynomials and top Lascoux polynomials*.
Algebraic Combinatorics (2025).

[11] *Top-degree components of Grothendieck and Lascoux polynomials*, with Pan, J.
Algebraic Combinatorics (2024).

[12] *A row analogue of Hecke column insertion*, with Huang, D. & Shimozono, M.
Combinatorial Theory (2024).

[13] *A bijection between K-Kohnert diagrams and reverse set-valued tableaux*, with Pan, J.
The Electronic Journal of Combinatorics (2023).

[14] *Set-valued tableaux rule for Lascoux polynomials*.
Combinatorial Theory (2023).

[15] *Grothendieck to Lascoux expansions*, with Shimozono, M.
Transactions of the American Mathematical Society (2023).

[16] *Harmonic bases for generalized coinvariant algebras*, with Rhoades, B. & Zhao, Z.
The Electronic Journal of Combinatorics (2020).

Conference Proceedings

[17] *Grothendieck positivity for square root crystals*, with Marberg, E. & Tong, K. H.
Proceedings of the 38th Conference on Formal Power Series and Algebraic Combinatorics, extended abstract of [2].

[18] *Grothendieck polynomials of fireworks and inverse fireworks permutations*, with Chou, J. C.-A. & Setiabrata, L.
Proceedings of the 38th Conference on Formal Power Series and Algebraic Combinatorics, extended abstract partially based on [6].

[19] *Generalized orthodontia, punched tableaux, and Lascoux expansion of sheared characters*, with St. Dizier, A. & Setiabrata, L.
Proceedings of the 38th Conference on Formal Power Series and Algebraic Combinatorics, extended abstract.

[20] *Marked bumpless pipedreams and compatible pairs*, with Huang, D. & Shimozono, M.
Proceedings of the 37th Conference on Formal Power Series and Algebraic Combinatorics, extended abstract of [4].

[21] *Analogues of two classical pipedream results on bumpless pipedreams*.
Proceedings of the 36th Conference on Formal Power Series and Algebraic Combinatorics, extended abstract of [5].

[22] *Grothendieck to Lascoux expansions*, with Shimozono, M.
Proceedings of the 34th Conference on Formal Power Series and Algebraic Combinatorics, extended abstract of [15].

[23] *Harmonic bases for generalized coinvariant algebras*, with Rhoades, B. & Zhao, Z.
Proceedings of the 33rd Conference on Formal Power Series and Algebraic Combinatorics, extended abstract of [16].

Awards and Grants

University of California, San Diego

- Selma and Robert Silagi Award for Undergraduate Excellence (5000 USD)

TALKS

A positive combinatorial formula for the double Edelman–Greene coefficients

- Rutgers University. New Brunswick, NJ. (invited) Sep. 2026
- University at Albany. Albany, NY. (invited) Sep. 2026
- University of Michigan. Ann Arbor, MI. Apr. 2026
- University of California, San Diego. La Jolla, CA. Feb. 2026
- University of California, Los Angeles. Los Angeles, CA. Feb. 2026
- Cornell University. Ithaca, NY. Jan. 2026
- Center for Combinatorics, Nankai University. Tianjin, China. Jan. 2026
- Sichuan University. Chengdu, China. (invited) Dec. 2025
- Beijing International Center for Mathematical Research. Beijing, China. Dec. 2025
- Canadian Mathematical Society Winter Meeting. Toronto, ON, Canada. Dec. 2025

An insertion algorithm for Schubert Cauchy identity via Pieri formula

- Geometry Seminar. Texas A&M University.College Station, TX. Mar. 2025
- Combinatorics Seminar. University of California, San Diego. La Jolla, CA. Mar. 2025

Grothendieck positivity for normal square root crystals

- Formal Power Series and Algebraic Combinatorics conference. University of Washington. Seattle, WA. Jul. 2026
- Canadian Mathematical Society Summer Meeting. Université Laval. Québec, QC, Canada. Jun. 2025
- AMS Sectional Meeting. University of Kansas. Lawrence, KS. Mar. 2025
- Richard P. Stanley Seminar in Combinatorics. Harvard University. Cambridge, MA. Feb. 2025
- Combinatorics Seminar. University of Florida. Gainesville, FL. Jan. 2025
- Combinatorics Seminar. Laboratoire d'Algèbre, de Combinatoire et d'Informatique Mathématique. Montréal, QC, Canada. Dec. 2024

K-theoretic analogue of Demazure crystal

- Hong Kong University of Science and Technology. Hong Kong, China. Sep. 2024

Three combinatorial formulas of Lascoux polynomials

- Sun Yat-sen University. Guangzhou, China. Aug. 2024
- Guangdong Technion-Israel Institute of Technology. Shantou, China. Aug. 2024

Embedding bumpless pipedreams as Bruhat chains

- Formal Power Series and Algebraic Combinatorics conference. Bochum, Germany. Jul. 2024
- Center for Combinatorics, Nankai University. Tianjin, China. Jul. 2024
- Beijing International Center for Mathematical Research. Beijing, China. Jul. 2024
- Combinatorics And Representation Theory Seminar. Seoul, Korea (via zoom). (invited) Apr. 2024
- Graduate Student Combinatorics Conference. Pittsburgh, PA. Mar. 2024
- The CAGE (Combinatorics, Algebra, and Geometry) Seminar. Philadelphia, PA. Mar. 2024
- Combinatorics Seminar. University of Southern California. Los Angeles, CA. (invited) Jan. 2024
- Combinatorial Algebra meets Algebraic Combinatorics. Montréal, QC, Canada. Jan. 2024
- Combinatorics Seminar. Laboratoire d'Algèbre, de Combinatoire et d'Informatique Mathématique. Montréal, QC, Canada. Dec. 2023
- Combinatorics Seminar. University of California, San Diego. La Jolla, CA. Nov. 2023
- Algebra Seminar. Virginia Polytechnic Institute and State University. Blacksburg, VA. Nov. 2023
- Combinatorics Seminar. University of Waterloo. Waterloo, ON, Canada. Sep. 2023
- Algebraic Combinatorics Seminar. Fields Institute. Toronto, ON, Canada. Sep. 2023

Pipedreams, bumpless pipedreams and Bruhat chains

- Schubert Seminar. Online. Nov. 2023

Top-degree components of Grothendieck and Lascoux polynomials

- Sichuan University. Chengdu, China (via zoom). (invited) Nov. 2024
- Combinatorics Seminar. University of Michigan. Ann Arbor, MI. (invited) Mar. 2023

- Graduate Student Combinatorics Conference. St. Louis, MO. Mar. 2023
- Combinatorics Seminar. University of California, San Diego. La Jolla, CA. Jan. 2023
- Combinatorics Seminar. University of California, Los Angeles. Los Angeles, CA. Nov. 2022

A row analogue of Hecke column insertion

- Combinatorics Seminar. University of Florida. Gainesville, FL. Nov. 2022
- Combinatorics Seminar. University of Minnesota. Minneapolis, MN. (invited) Sep. 2022
- Combinatorics Seminar. University of Illinois, Urbana-Champaign. Urbana, IL. Sep. 2022

Grothendieck to Lascoux expansions

- Combinatorics and Geometry Seminar. University of Washington. Seattle, WA. Apr. 2022
- Graduate Student Combinatorics Conference. San Diego, CA (via zoom). Mar. 2022
- Algebra & Combinatorics Seminar. North Carolina State University. Raleigh, NC. (invited) Nov. 2021
- Algebra Seminar. Virginia Polytechnic Institute and State University. Blacksburg, VA. Nov. 2021
- Combinatorics Seminar. University of California, San Diego. La Jolla, CA. Oct. 2021
- Combinatorics Seminar. University of Waterloo. Waterloo, ON, Canada (via zoom). (invited) Sep. 2021
- Combinatorics Seminar. University of Illinois, Urbana-Champaign. Urbana, IL. (invited) Sep. 2021
- Combinatorics Seminar. University of Minnesota. Minneapolis, MN. Sep. 2021

Harmonic bases for generalized coinvariant algebras

- Formal Power Series and Algebraic Combinatorics conference. Ramat-Gan, Israel (via zoom). Jan. 2022
- Combinatorics Seminar. University of California, San Diego. La Jolla, CA Jun. 2021

Teaching Experience

Department of Mathematics, UCSD

Teaching Assistant: Calculus, Mathematical Reasoning, Combinatorics, Algebraic Combinatorics.

Department of Computer Science and Engineering, UCSD

Teaching Assistant: Discrete Mathematics, Design and Analysis of Algorithms, Theory of Computability.